
Failure of Preenucleation Radiation to Decrease Uveal Melanoma Mortality

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We analyzed uveal melanoma metastases in a group of 41 patients who received 20 Gy of preenucleation radiation in a Northern California Oncology Group preliminary phase I/II study, and compared their survival rates with a retrospective control group of 31 patients with characteristics matching the entrance criteria but treated with enucleation alone. Using the Cox proportional hazards model, we found that increased tumor diameter, mixed or epithelioid cell type, and radiation adversely affected survival. In vivo studies of cell cycling indicated that 20 Gy of preenucleation radiation appeared to diminish the reproductive integrity of the tumor cells. It is most likely that the failure of preenucleation irradiation to prolong patient survival was because of micrometastases that occurred before treatment.

WHILE THE OPTIMUM management of large uveal melanomas is uncertain,¹ these tumors have been treated with enucleation alone, perioperative irradiation with enucleation, or different forms of radiation with intent to retain the globe. Possible adverse effects of enucleation trauma have been hypothesized to increase melanoma-related mortality.² In some other body sites, perioperative irradiation appears to decrease tumor-related mortality.³

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Conceivably, if trauma at the time of enucleation was associated with increased tumor mortality, preenucleation radiation might decrease it. In 1977 we initiated a Northern California Oncology Group (NCOG) preliminary phase I/II study of 20 Gy of preenucleation photon irradiation in patients with large uveal melanomas to study some of these issues.³

See also p. 88.

Herein we used a Cox proportional hazards model to analyze retrospectively uveal melanoma metastases after 20 Gy of preenucleation irradiation and observed that this therapy does not favorably affect tumor-related mortality.

Material and Methods

All patients were examined in the Ocular Oncology Unit, University of California, San Francisco, between 1976 and 1985. The diagnosis of a uveal melanoma was established based on multimodality, multiobserver techniques including indirect ophthalmoscopy, fundus photography, fluorescein angiography, and ultrasonography.⁴ All patients had a complete physical examination before surgery; the metastatic evaluation included a chest x-ray and serum lactic dehydrogenase, alkaline phosphatase, and glutamyl transpeptidase levels. In patients with abnormal physical or laboratory findings, further appropriate metastatic studies including body computed tomography and fine needle aspiration biopsies were performed. No patient had known metastatic disease before treatment.

Patients treated as part of a phase I/II NCOG study of preenucleation irradiation gave both written and oral informed consent before therapy. The criteria for inclusion in the radiation study were a uveal melanoma that was greater than 15 mm in largest diameter or greater than